

1. According to the Absorption Law, what does the expression $A \cdot (A + B)$ simplify to?

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|-------------|-----------------|
| (A) A | (B) B |
| (C) $A + B$ | (D) $A \cdot B$ |

2. What is the output of a 2-input NAND gate when both inputs are set to 1?

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|--------------------|---------------|
| (A) 0 | (B) 1 |
| (C) High Impedance | (D) Undefined |

3. According to the Associative Law of Addition, $(A + B) + C$ is equal to:

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|---------------------|-----------------------|
| (A) $A + (B + C)$ | (B) $A \cdot (B + C)$ |
| (C) $A + B \cdot C$ | (D) $(A \cdot B) + C$ |

4. What is the value of the Boolean expression $1 + A + B$?

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|-------|-------------------|
| (A) 1 | (B) $A + B$ |
| (C) 0 | (D) $A' \cdot B'$ |

5. Which logic gate produces an output of 1 when an odd number of inputs are 1?

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|---------|----------|
| (A) XOR | (B) XNOR |
| (C) NOR | (D) NAND |

6. What is the dual of the Boolean statement $A \cdot 1 = A$?

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|------------------|---------------------|
| (A) $A + 0 = A$ | (B) $A \cdot 0 = 0$ |
| (C) $A' + 1 = 1$ | (D) $A + 1 = A$ |

7. Simplify the expression: $(A + B)' + (A + B)$.

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|-------------|-------------------|
| (A) 1 | (B) 0 |
| (C) $A + B$ | (D) $A' \cdot B'$ |

8. How many minterms exist for a Boolean function with 4 variables?

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|--------|--------|
| (A) 16 | (B) 8 |
| (C) 4 | (D) 32 |

9. According to the second distributive law in Boolean algebra, $A + (B \cdot C)$ is equivalent to:

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|-----------------------------|---------------------|
| (A) $(A + B) \cdot (A + C)$ | (B) $A \cdot B + C$ |
| (C) $A \cdot B + A \cdot C$ | (D) $A + B + C$ |

10. What is the complement of the expression $A \cdot B'$?

- (A) $A' + B$
- (B) $A' \cdot B$
- (C) $A + B'$
- (D) $A' + B'$

11. What is the output of an XNOR gate when both inputs are equal?

- (A) 1
- (B) 0
- (C) Inverted Input
- (D) High Impedance

12. Besides NAND, which other standard logic gate is functionally complete (universal)?

- (A) NOR
- (B) XOR
- (C) AND
- (D) OR

13. Simplify the expression: $A \cdot (A' + B)$.

- (A) $A \cdot B$
- (B) $A + B$
- (C) $A' \cdot B$
- (D) 0

14. What is the complement of constant 1 in Boolean algebra?

- (A) 0
- (B) 1
- (C) -1
- (D) Undefined

15. A minterm is a product term that contains:

- (A) All variables in complemented or uncomplemented form
- (B) Only complemented variables
- (C) At least one OR operation
- (D) Only constant values

16. What is the product of maxterms representation also known as?

- (A) Conjunctive Normal Form (CNF)
- (B) Disjunctive Normal Form (DNF)
- (C) Sum of Products (SOP)
- (D) Minimal Cover

17. Simplify the expression: $A \cdot B + A \cdot B'$.

- (A) A
- (B) B
- (C) $A + B$
- (D) 0

18. Which law allows rearranging terms like $A \cdot B = B \cdot A$?

- (A) Commutative Law
- (B) Associative Law
- (C) Distributive Law
- (D) Involution Law

19. What is the output of an AND gate if one input is A and the other is 0?

- (A) 0 (B) A
(C) 1 (D) A'

20. What is the output of an OR gate if one input is A and the other is 0?

- (A) A (B) 0
(C) 1 (D) A'

21. Simplify the expression: $A + A'$.

- (A) 1 (B) 0
(C) A (D) A'

22. What logic gate corresponds to the expression $A' \cdot B'$?

- (A) NOR (B) NAND
(C) OR (D) AND

23. What logic gate corresponds to the expression $A' + B'$?

- (A) NAND (B) NOR
(C) XOR (D) XNOR

24. According to the Consensus Theorem, what does $A \cdot B + A' \cdot C + B \cdot C$ simplify to?

- (A) $A \cdot B + A' \cdot C$ (B) $B \cdot C$
(C) $A \cdot B + B \cdot C$ (D) $A' \cdot C + B \cdot C$

25. Simplify the expression: $A \cdot (A + B + C + D)$.

- (A) A (B) $A + B + C + D$
(C) $A \cdot B \cdot C \cdot D$ (D) 0

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)